



A Darboux transformation for the Volterra lattice equation

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Abstract

A Darboux transformation is presented for the Volterra lattice equation, based on a pair of 2×2 matrix spectral problems. The resulting DT is applied to construction of solitary wave solutions from a constant seed solution. A particular phenomenon is that only one condition is required in determining the corresponding Darboux matrix, but not two as for most pairs of 2×2 matrix spectral problems.

Keywords Matrix spectral problem · Integrable lattice equation · Darboux transformation

Mathematics Subject Classification 35Q51 · 35Q53 · 37K40

1 Introduction

The Darboux transformation (DT) is one of the powerful methods to solve nonlinear integrable equations [1,2]. The starting point is a pair of matrix spectral problems (or a Lax pair [3]), which plays a crucial role in solving nonlinear integrable equations in the inverse scattering transform [4,5]. There are many other powerful theories in modern soliton theory, including Riemann-Hilbert problems [5], Sato τ -function theory [6],

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and isomonodromic deformations [7], all of which underline the importance of matrix spectral problems (or Lax pairs).

Compatibility conditions of pairs of matrix spectral problems generating nonlinear integrable equations, in both continuous and discrete cases, are also called zero curvature equations, whose Liouville integrability could be explored through the trace identities [8,9] or the variational identities [10,11]. The basis of the DT theory [1,2] is a sort of gauge transformations, which leave zero curvature equations invariant, and the whole process leads from an old solution to a new one (see, e.g., [12,13]).

We will discuss lattice equations within the zero curvature formulation [9,11]. Let f be a lattice function. The shift operator E and the reverse shift operator E^{-1} are defined by

$$(Ef)(n) = f(n+1), \quad (E^{-1}f)(n) = f(n-1), \quad n \in \mathbb{Z}, \quad (1.1)$$

and the iteration of the shift or the reverse shift is simply written as

$$f^{(k)} = E^k f, \quad k \in \mathbb{Z}. \quad (1.2)$$

A lattice equation

$$u_t = K(u, u^{(1)}, u^{(-1)}, \dots), \quad u = u(n, t), \quad (1.3)$$

is said to possess a Lax pair, U and V , if it can be expressed as a compatibility condition (i.e., a discrete zero curvature equation, see, e.g., [9,11])

$$U_t = (EV)U - UV \quad (1.4)$$

of a spatial matrix discrete spectral problem

$$E\phi = U(u, \lambda)\phi \quad (1.5)$$

and an associated temporal matrix discrete spectral problem

$$\phi_t = V(u, u^{(1)}, u^{(-1)}, \dots; \lambda)\phi, \quad (1.6)$$

where λ is a spectral parameter, ϕ is an eigenfunction, and U is called a spectral matrix and V , a Lax operator or matrix. In the setting up of the Lax theory, the key is a spatial matrix spectral problem. Usually, the equations resulting from matrix spectral problems carry high nonlinearity, but they possess Abelian Lax operator algebras [14] and a generating function for conserved functionals can be constructed from the corresponding discrete trace identity [9] or discrete variational identity [11]. Therefore, infinitely many symmetries and infinitely many conserved functionals can be guaranteed, which exhibit Liouville integrability of the equations under consideration.

In this paper, we would like to present a Darboux transformation for the Volterra lattice equation from a pair of 2×2 matrix spectral problems, which is different from a DT generated from a pair of scalar spectral problems in [1]. The basic technique to be adopted will be the discrete gauge transformation technique (see, e.g., [15–17]). Unlike the standard case of 2×2 matrix discrete spectral problems [18,19], the Volterra

lattice equation will only need a special condition in the construction of the Darboux transformation. A few concluding remarks will be given in the last section.

2 Darboux transformation

We consider the Volterra lattice equation [20]:

$$u_t = u(u^{(-1)} - u^{(1)}), \quad (2.1)$$

where $u = u(n, t)$. This Volterra equation has various applications, particularly in mathematical biology [20]. Our starting point is a spatial 2×2 matrix discrete spectral problem:

$$E\phi = U\phi, \quad U = U(u, \lambda) = \begin{bmatrix} \lambda & \lambda u \\ 1 & 0 \end{bmatrix}, \quad (2.2)$$

where u is the potential in (2.1) and λ is the spectral parameter. The spectral problem (2.2) is a scalar multiple of the spectral problem in [14]. It is direct to see that the Volterra lattice equation (2.1) can be expressed as the compatibility condition of the spatial matrix discrete spectral problem (2.2) and the temporal matrix spectral problem

$$\phi_t = V\phi, \quad V = V(u, \lambda) = \begin{bmatrix} \frac{1}{2}\lambda - u & \lambda u \\ 1 & -\frac{1}{2}\lambda - u^{(-1)} \end{bmatrix}, \quad (2.3)$$

which was presented in [14].

A Darboux transformation is usually generated by a so-called Darboux matrix $T = T(u, \lambda)$, which satisfies the conditions

$$T^{(1)}U = \tilde{U}T \quad (2.4)$$

and

$$T_t T + TV = \tilde{V}T. \quad (2.5)$$

In the above computations, we demand

$$\tilde{U} = U(\tilde{u}, \lambda) = \begin{bmatrix} \lambda & \lambda \tilde{u} \\ 1 & 0 \end{bmatrix}, \quad \tilde{V} = V(\tilde{u}, \lambda) = \begin{bmatrix} \frac{1}{2}\lambda - \tilde{u} & \lambda \tilde{u} \\ 1 & -\frac{1}{2}\lambda - \tilde{u}^{(-1)} \end{bmatrix}. \quad (2.6)$$

Therefore, when T is invertible, we have a new Lax pair $(\tilde{U}, \tilde{V}) = (T^{(1)}UT^{-1}, T_t + TVT^{-1})$ and a new pair of matrix spectral problems

$$E\tilde{\phi} = \tilde{U}\tilde{\phi}, \quad \tilde{\phi}_t = \tilde{V}\tilde{\phi}, \quad \tilde{\phi} = T\phi, \quad (2.7)$$

which present the same zero curvature equation leading to the Volterra lattice equation (2.1).

We would like to find a Darboux matrix of the following form

$$T = T(u, \lambda) = \begin{bmatrix} \lambda A & \lambda B \\ C & \lambda D \end{bmatrix}. \quad (2.8)$$

For such a matrix T , the spatial condition (2.4) requires

$$\begin{cases} C^{(1)}u = B, \\ A = C^{(1)} + D^{(1)}, \\ -A^{(1)}u + D\tilde{u} + B = 0, \\ A = A^{(1)}, \\ C\tilde{u} - B^{(1)} = 0. \end{cases} \quad (2.9)$$

We would like to determine a set of differential functions A , B , C and D which satisfy (2.9), and thus, generate a new solution $\tilde{u}$.

As usual, we take a special eigenfunction associated with an eigenvalue λ_0 :

$$E\Gamma = U(u, \lambda_0)\Gamma, \quad \Gamma_t = V(u, \lambda_0)\Gamma, \quad (2.10)$$

where $\Gamma = (\Gamma_1, \Gamma_2)^T$ and u is a given solution of (2.1). Then, upon setting

$$\sigma = \frac{\Gamma_2}{\Gamma_1}, \quad (2.11)$$

we can easily find from (2.10) that

$$E\sigma = \frac{1}{\lambda_0(1 + u\sigma)}, \quad \sigma_t = 1 - (\lambda_0 - u + u^{(-1)})\sigma - \lambda_0 u \sigma^2. \quad (2.12)$$

Let us require one condition $T(\lambda_0)\Gamma = 0$, and so, we have

$$\begin{cases} \lambda_0 A + \lambda_0 B\sigma = 0, \\ C + \lambda_0 D\sigma = 0. \end{cases}$$

Together with the second and fourth equations in (2.9), this leads to

$$A = -\alpha, \quad B = \frac{\alpha}{\sigma}, \quad C = -\lambda_0 D\sigma, \quad D = -\frac{\alpha}{1 - \lambda_0 \sigma}, \quad (2.13)$$

where α is an arbitrary constant. The first equation in (2.9) is a consequence of the first equation in (2.12), and the third equation in (2.9) is automatically satisfied with the help of the fifth equation in (2.9). Therefore, all differential functions for the Darboux matrix T are determined through

$$A = -\alpha, \quad B = \frac{\alpha}{\sigma}, \quad C = \frac{\alpha\lambda_0\sigma}{1 - \lambda_0\sigma}, \quad D = -\frac{\alpha}{1 - \lambda_0\sigma}, \quad \alpha = \text{const.} \neq 0, \quad (2.14)$$

and the new solution is given by

$$\tilde{u} = \frac{B^{(1)}}{C} = \frac{(1 - \lambda_0 \sigma)(1 + u \sigma)}{\sigma}. \quad (2.15)$$

We can directly prove that such an invertible Darboux matrix also satisfies (2.5), by using the second equation in (2.12). A Maple symbolic computation for this is simpler. Such an amazing coherence often occurs for integrable equations.

Now, all above computations show that the resulting Darboux transformation of the Volterra lattice equation reads

$$\{u, U, V\} \rightarrow \{\tilde{u}, \tilde{U}, \tilde{V}\} = \{(1 - \lambda_0 \sigma)(u + \sigma^{-1}), T^{(1)} U T^{-1}, T_t + T V T^{-1}\} \quad (2.16)$$

where σ is determined by (2.11) with (2.10) and T is defined by (2.8) with (2.14). By iterating this transformation, one is able to generate exact solutions involving as many eigenfunctions as one wishes.

Let us finally apply the resulting DT to generate solitary wave solutions from a constant seed solution $u = \mu_0 = \text{const}$. At this moment, we have

$$E \Gamma_1 = \lambda_0 \Gamma_1 + \lambda_0 \mu_0 \Gamma_2, \quad E \Gamma_2 = \Gamma_1, \quad (2.17)$$

and

$$\begin{cases} \Gamma_{1,t} = \left(\frac{1}{2}\lambda_0 - \mu_0\right) \Gamma_1 + \lambda_0 \mu_0 \Gamma_2, \\ \Gamma_{2,t} = \Gamma_1 - \left(\frac{1}{2}\lambda_0 + \mu_0\right) \Gamma_2. \end{cases} \quad (2.18)$$

Introduce

$$\lambda_{\pm} = \frac{\lambda_0 \pm \sqrt{\lambda_0^2 + 4\lambda_0 \mu_0}}{2}, \quad (2.19)$$

and then the general solution of the linear difference equation (2.17) reads

$$\Gamma_1 = v_1 \lambda_+^{n+1} + v_2 \lambda_-^{n+1}, \quad \Gamma_2 = v_1 \lambda_+^n + v_2 \lambda_-^n. \quad (2.20)$$

Plugging this solution into the linear differential equation (2.18) engenders

$$\begin{cases} v_{1,t} = \left(\lambda_+ - \frac{1}{2}\lambda_0 - \mu_0\right) v_1, \\ v_{2,t} = \left(\lambda_- - \frac{1}{2}\lambda_0 - \mu_0\right) v_2. \end{cases} \quad (2.21)$$

Then, we can immediately obtain

$$\Gamma_1 = \Gamma_2^{(1)}, \quad \Gamma_2 = v_{1,0} e^{(\lambda_+ - \frac{1}{2}\lambda_0 - \mu_0)t} \lambda_+^n + v_{2,0} e^{(\lambda_- - \frac{1}{2}\lambda_0 - \mu_0)t} \lambda_-^n, \quad (2.22)$$

where $v_{1,0}$ and $v_{2,0}$ are arbitrary constants, and further, the new solution by the DT above is

$$\tilde{u} = (\sigma^{-1} - \lambda_0)(1 + \mu_0\sigma), \quad \sigma = \frac{\Gamma_2}{\Gamma_1}, \quad (2.23)$$

where Γ_1 and Γ_2 are given by (2.22).

3 Concluding remarks

We have presented a Darboux transformation (DT) for the Volterra lattice equation from a pair of 2×2 matrix spectral problems, and applied the resulting DT to the construction of solitary wave solutions of the Volterra lattice equation. Our DT is different from a Darboux transformation presented from a pair of scalar spectral problems in [1]. A particular phenomenon is that we have used only one condition, $T(u, \lambda_0)\Gamma = 0$, in determining the Darboux matrix T . This is partially due to the fact that there exists only one potential in the spatial discrete spectral problem. The specific dependence of the Darboux matrix T on λ in (2.8) is very crucial. The obtained DT is just one small flower that grows in a land of integrable equations, which blooms here at this moment.

We point out that there are recent studies on lump solutions to continuous integrable equations in (2+1)-dimensions (see, e.g., [21–23]) and linear partial differential equations in (3+1)-dimensions (see, e.g., [24]). There are even more interaction solutions between lumps and other kinds of exact solutions in (2+1)-dimensions (see, e.g., [25–28] for lump-kink interaction solutions and [29–31] for lump-soliton interaction solutions). The existence of soliton solutions and interaction solutions to nonlinear equations characterizes their complete integrability, from which symmetries and conserved quantities could be derived. All such studies enrich the existing theories of soliton solutions and dromion-type solutions, through various efficient approaches such as the Hirota perturbation technique, symmetry reductions, symmetry constraints and the Riemann-Hilbert technique (see, e.g., [32–39]).

It will be very interesting and important to generate soliton and interaction solutions and study their patterns for integrable lattice equations, and to identify nonlinear lattice equations that possess Darboux transformations and soliton and interaction solutions.

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Compliance with ethical standards

Conflict of interest The author declares that there is no conflict of interest.

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